Project Proposal: Explaining Patient-Similarity Models on MIMIC-IV

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Summary

We propose to explain a **patient-similarity model** used for clinical-trial cohort matching: given a diabetes patient from MIMIC-IV, the model returns the k most similar patients. We compare two models, a count-based KNN baseline and KNN on embeddings from the pretrained EHR foundation model **CLMBR-T-base**. Phase I tests whether the neighbors are clinically meaningful and fair; Phase II opens the embedding space to see what the model actually encodes. The idea comes from real startup practice, where KNN is used to match patients to trials.

Moodle registration entry (due 28.10, 23:59)

One member submits this for the whole group; it follows the format of the "ideal entry" shown in the Phase I examples session. Fill in the group name and student IDs before submitting.

Task

We investigate patient-similarity retrieval for clinical-trial cohort matching, using diabetes patients from the MIMIC-IV electronic health record database. Given a patient, the model returns the 10 most similar patients; a good neighbor set shares the patient's later complication pathway (kidney, cardiovascular, eye, or none).

Our goal is to explore model behavior through visual analysis of its inputs and outputs, identifying patterns such as errors, demographic biases and sensitivity to small input changes. Building on these findings, we will examine the models' internal patient representations to understand what they learned and how this explains their behavior.

Dataset: MIMIC-IV

Paper: EHRSHOT (Wornow et al., NeurIPS 2023)

Model

We use CLMBR-T-base, a 141M-parameter transformer foundation model pretrained on structured EHRs of 2.57M patients, and run k-nearest-neighbor search on its patient embeddings. We compare it with a count-based KNN baseline on engineered clinical features.

If access to the pretrained weights is delayed, we will replace CLMBR-T-base with Bio_ClinicalBERT embeddings of discharge notes.

Repository: ehrshot-benchmark

Model weights: StanfordShahLab/clmbr-t-base

Task

Patient-similarity retrieval for diabetes complication cohorts. Input: one patient's record (diagnoses, labs, medications, admissions). Output: the k = 10 most similar patients. A neighbor set is good when its patients share the query patient's later complication pathway: kidney, cardiovascular, eye, or none.

Why this task fits the assignment:

- It is well-defined, with a clear ground truth (later diagnoses), so failures are measurable.
- It is a real use case: trial recruitment depends on finding comparable patients, and bad neighbors mean biased trial cohorts.
- It gives us two models to compare, which the assignment explicitly allows.
- It maps directly onto the course topics: projections, embeddings and trajectories (Lecture 3).

Data

Main dataset: MIMIC-IV (Beth Israel Deaconess Medical Center, Boston, about 2008–2019). We use the `hosp` module: `diagnoses_icd`, `labevents` (HbA1c, creatinine/eGFR, glucose, lipids), `prescriptions` and `admissions`, linked per patient by `subject_id`. Cohort: adults with a diabetes diagnosis (ICD-9 250.x, ICD-10 E10–E14) and at least two admissions, so later complications can be observed.

Dataset	Use	Access
MIMIC-IV	Main analysis	PhysioNet credentialing + data use agreement, each member
MIMIC-IV OMOP Demo (100 patients)	Build and test the pipeline before approval	Free, no registration
UCI Diabetes 130-US Hospitals	Fallback if access is delayed	Public

Access plan: all four members complete the CITI course "Data or Specimens Only Research" and apply on PhysioNet this week. Approval usually takes several business days (PhysioNet FAQ). Known limitations: hospital-only data, so outpatient progression is missing and eye disease is underrepresented.

Models

We compare a simple baseline with a pretrained foundation model, both feeding the same KNN step.

Model	How it represents a patient	Why we chose it
A. Count-based KNN	Vector of code counts (diagnoses, labs, drugs) plus age	Transparent baseline; every distance can be traced to features
B. CLMBR-T-base + KNN	768-dimensional embedding from a 141M-parameter transformer pretrained on 2.57M Stanford patient records	Learned representation with real internals to open in Phase II

Why CLMBR-T-base is the best fit: it works on structured EHR codes (not text), produces patient embeddings directly usable for KNN, and comes with open benchmark code (EHRSHOT) that already includes a count-based baseline. It was pretrained on a different hospital, so it may transfer imperfectly to MIMIC-IV, which is itself an interesting finding to test.

Requirements: MIMIC-IV must be converted to OMOP (open-source OHDSI ETL), and the weights need a Stanford non-commercial data use agreement. That is fine for the course but rules out reuse in a commercial product.

Alternatives considered:

- EHRMamba: trained and evaluated on MIMIC-IV itself, but likely needs our own GPU training via the Odyssey toolkit.
- Bio_ClinicalBERT: easiest setup (MIT license, two lines of code) but uses clinical notes only. **This is our fallback** if the OMOP conversion or the Stanford agreement blocks us.

Phase I: how do the models behave?

Black box closed: we only look at which patients go in and which neighbors come out. Each question gets an investigative visualization, not a single accuracy chart.

We build on the three investigative tools from the Phase I practical-examples session. In Phase I, any projection uses **input features only** (diagnoses, labs, drugs), never the model's embeddings; those belong to Phase II.

1. Scatter plot with one variable per visual channel. Each dot is a query patient: x = neighbor agreement (share of the 10 neighbors with the same later complication), y = HbA1c or eGFR, color = sex or ethnicity, shape = insurance type, small multiples = complication pathway. This can reveal statements like "patients with low eGFR get the least consistent neighbors" or "agreement is lower for one ethnic group."

2. Confusion matrix of predicted vs. actual pathway. We predict each patient's pathway by majority vote of their neighbors, then reorder rows and columns by error similarity to reveal blocks. If kidney and cardiovascular patients are mixed up with each other but not with eye patients, the models have learned a clinically sensible structure; random confusions mean they have not. We compare both models' matrices side by side.

3. Projection of patients with drill-down. A UMAP of input features, colored by neighbor agreement, shows where in patient space the models fail. We drill down to individual failing patients, summarize each failure region and annotate it, as in the Projection Space Explorer approach.

4. Paths over admissions. Each patient's admissions form a sequence, so we connect them into a path in the projection, as in the Projection Path Explorer example. The path shows how a patient's neighbor set and predicted pathway change as their disease progresses.

Synthetic inputs to test hypotheses: perturbed copies of real records (one value changed, such as HbA1c 7% → 9%) and counterfactuals (same record, different sex or age). A "neighbor flip" chart shows how many neighbors change per edit. Interpretable attributes we derive: comorbidity count, record completeness, number of admissions.

Question	Pattern type	Tool
Do neighbors share the later complication?	Errors	Scatter plot, confusion matrix
Are neighbors chosen by demographics rather than clinical state?	Bias	Scatter plot color channel, counterfactuals
How sensitive are neighbors to one changed lab value?	Output shift	Perturbations, neighbor-flip chart
Do sparse records get meaningless neighbors?	Edge cases	Scatter plot vs. record completeness

Question	Pattern type	Tool
Do models A and B agree?	Comparison	Side-by-side confusion matrices, overlap matrix

Phase II: what did the models learn?

Black box open: we test the Phase I findings against the models' internal representations.

- **Projections:** UMAP of both patient spaces, colored first by complication, then by hospital unit and admission year. If model B clusters by year rather than disease, it has learned a shortcut.
- **Distance decomposition (model A):** which features contribute most to each neighbor distance, to explain the Phase I flips.
- **Probing (model B):** small linear classifiers on the embeddings test what is encoded, such as age, sex, kidney function or insurance. Strong demographic signal would explain any Phase I bias.
- **Trajectories:** each patient's admissions drawn as a path in embedding space, showing how their neighbors change as disease progresses.

Each Phase II view links back to one Phase I observation, so the final story reads: what we saw, then why it happens.

AI4VIS: using generative AI

Two members design with AI help and two without in each phase, then we swap, as the assignment requires. We record key prompts and which suggestions we accepted, rejected or refined, and why.

Data-safety rule: the MIMIC data use agreement forbids sending data to third-party online services. We never paste real patient rows into AI tools; we use them only for design ideas, code, and synthetic or aggregated examples. We will report this constraint and how it shaped our AI use as a best-practice finding.

GitHub repository

The course requires a **public** GitHub repository for our code, with two files in its root, filled in before submission:

- `team-info.json`: team name and, per student, student ID, first and last name, email and GitHub username.

- `workload.json`: team name and, per student, first and last name and workload share (e.g. 0.25 each for an even split).

Data-safety rule: because the repository is public, it must never contain MIMIC-IV data, extracted cohorts, embeddings, neighbor lists or any patient-level output. We add a `.gitignore` for all data folders from day one, commit only code and aggregated figures, and use the free MIMIC-IV demo or synthetic patients for any example shown in the repository or presentation.

Expected outcomes

We expect to deliver an interactive visual analysis comparing both models, plus findings we can state with evidence. Hypotheses we will test:

1. The pretrained model B finds more clinically consistent neighbors than baseline A for kidney and cardiovascular pathways.
2. Both models perform worse for eye complications, because retinopathy is rare in hospital data.
3. Some neighbor choices are driven by demographics or admission year rather than disease, which would bias trial cohorts.
4. Model B's embedding encodes Stanford-specific patterns that transfer imperfectly to MIMIC-IV.

A negative result on any hypothesis is still a valid finding. Beyond the course, the same analysis shows how to audit patient-matching tools before they are used for real trial recruitment.

Risks and mitigations

Risk	Mitigation
MIMIC-IV approval is slow for one or more members	Apply this week; build the pipeline on the free OMOP demo meanwhile
Stanford agreement for CLMBR-T-base is delayed or refused	Switch model B to Bio_ClinicalBERT on discharge notes
OMOP conversion takes too long	Use the OHDSI ETL as is; limit features to diagnoses, labs, drugs
Too few diabetes patients with repeat admissions	Widen the cohort to any admission with a diabetes code

Risk	Mitigation
Not enough compute	Inference only for CLMBR-T-base; no pretraining needed

Timeline and task split

Model selection is due 28.10, the midpoint 11.11, the final deadline 09.12, followed by the mandatory presentation.

Due	Milestone	Work
28.10	Model selection registered	One member submits the Moodle entry; create the public GitHub repo; all start credentialing
04.11	Pipeline running	OMOP demo pipeline, both models producing neighbors
11.11	Midpoint: Phase I	Phase I visualizations on full MIMIC-IV
02.12	Phase II done	Projections, probing, trajectories
09.12	Final submission	Report, AI4VIS log, presentation

Proposed roles (names to assign at our next meeting):

- **Data lead:** MIMIC-IV cohort, OMOP conversion, features.
- **Model lead:** CLMBR-T-base embeddings, count baseline, KNN.
- **Phase I lead:** behavior analyses and visualizations.
- **Phase II lead:** projections, probing, trajectories.

Open question: who takes the AI4VIS log and the final presentation?

Key papers and links

- Wornow et al., EHRSHOT: An EHR Benchmark for Few-Shot Evaluation of Foundation Models, NeurIPS 2023. Source of CLMBR-T-base and the baseline code.
- Fallahpour et al., EHRMamba: Towards Generalizable and Scalable Foundation Models for EHRs, ML4H 2024. Alternative trained on MIMIC-IV.
- Alsentzer et al., Publicly Available Clinical BERT Embeddings, NAACL Clinical NLP Workshop 2019. Fallback text model.
- Yu et al., Benchmarking Foundation Models with Multimodal Public EHRs, 2025. MIMIC-IV preprocessing, fairness and interpretability reference.

- MIMIC-IV getting started and PhysioNet credentialing FAQ.